2 Vector and Calculus

Addition and Subtraction of Two Vectors

1. If two vectors $\vec{A}$ and $\vec{B}$ having equal magnitude R are inclined at an angle θ, then **[31 Jan, 2024 (Shift-II)]**

(a) $|\vec{A}-\vec{B}|=\sqrt{2}\,R\sin\frac{\theta}{2}$ (b) $|\vec{A}+\vec{B}|=2\,R\sin\frac{\theta}{2}$

(c) $|\vec{A}+\vec{B}|=2\,R\cos\frac{\theta}{2}$ (d) $|\vec{A}-\vec{B}|=2\,R\cos\frac{\theta}{2}$

2. A vector has magnitude same as that of $\vec{A}=3\hat{i}+4\hat{j}$ and is parallel to $\vec{B}=4\hat{i}+3\hat{j}$. The x and y components of this vector in first quadrant are x and 3 respectively where $x=$ ________. **[30 Jan, 2024 (Shift-II)]**

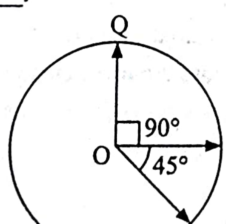
3. Two forces $\vec{F}_1$ and $\vec{F}_2$ are acting on a body. One force has magnitude thrice that of the other force and the resultant of the two forces is equal to the force of larger magnitude. The angle between $\vec{F}_1$ and $\vec{F}_2$ is $\cos^{-1}\left(\frac{1}{n}\right)$. The value of $|n|$ is ____. **[04 April, 2024 (Shift-I)]**

4. The angle between vector $\vec{Q}$ and the resultant of $(2\vec{Q}+2\vec{P})$ and $(2\vec{Q}-2\vec{P})$ is: **[05 April, 2024 (Shift-I)]**

(a) $0°$ (b) $\tan^{-1}\frac{(2\vec{Q}-2\vec{P})}{2\vec{Q}+2\vec{P}}$

(c) $\tan^{-1}\left(\frac{P}{Q}\right)$ (d) $\tan^{-1}\left(\frac{2Q}{P}\right)$

5. Three vectors $\overline{OP}, \overline{OQ}$ and $\overline{OR}$ each of magnitude A are acting as shown in figure. The resultant of the three vectors is $A\sqrt{x}$. The value of x is _____. **[08 April, 2024 (Shift-I)]**



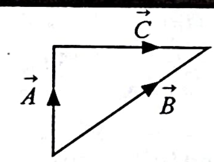
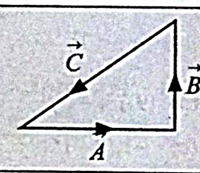
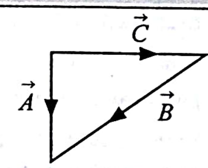
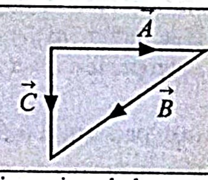
6. If $\vec{a}$ and $\vec{b}$ makes an angle $\cos^{-1}\left(\frac{5}{9}\right)$ with each other, then $|\vec{a}+\vec{b}|=\sqrt{2}\,|\vec{a}-\vec{b}|$ for $|\vec{a}|=n|\vec{b}|$. The integer value of n is ______. **[09 April, 2024 (Shift-I)]**

7. The resultant of two vectors $\vec{A}$ and $\vec{B}$ is perpendicular to $\vec{A}$ and its magnitude is half that of $\vec{B}$. The angle between vectors $\vec{A}$ and $\vec{B}$ is .... **[09 April, 2024 (Shift-II)]**

8. When vector $\vec{A}=2\hat{i}+3\hat{j}+2\hat{k}$ is subtracted from vector $\vec{B}$, it gives a vector equal to $2\hat{j}$. Then the magnitude of vector $\vec{B}$ will be: **[11 April, 2023 (Shift-II)]**

(a) $\sqrt{13}$ (b) 3 (c) $\sqrt{6}$ (d) $\sqrt{5}$

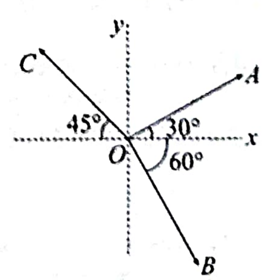
9. Match List-I with List-II.

	List-I		List-II
A.	$\vec{C}-\vec{A}-\vec{B}=0$	I.	
B.	$\vec{A}-\vec{C}-\vec{B}=0$	II.	
C.	$\vec{B}-\vec{A}-\vec{C}=0$	III.	
D.	$\vec{A}+\vec{B}=-\vec{C}$	IV.	

Choose the correct answer from the options given below: **[25 July, 2021 (Shift-I)]**

(a) A-IV, B-I, C-III, D-II (b) A-IV, B-III, C-I, D-II
(c) A-I, B-IV, C-II, D-III (d) A-III, B-II, C-IV, D-I

10. The magnitude of vectors $\overline{OA}, \overline{OB}$ and $\overline{OC}$ in the given figure are equal. The direction of $\overline{OA}+\overline{OB}-\overline{OC}$ with x-axis will be **[26 Aug, 2021 (Shift-I)]**



(a) $\tan^{-1}\frac{(\sqrt{3}-1+\sqrt{2})}{(1-\sqrt{3}+\sqrt{2})}$

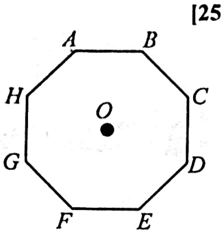
(b) $\tan^{-1}\frac{(1+\sqrt{3}-\sqrt{2})}{(1-\sqrt{3}-\sqrt{2})}$

(c) $\tan^{-1}\frac{(1-\sqrt{3}-\sqrt{2})}{(1+\sqrt{3}+\sqrt{2})}$

(d) $\tan^{-1}\frac{(\sqrt{3}-1+\sqrt{2})}{(1+\sqrt{3}-\sqrt{2})}$

11. In an octagon $ABCDEFGH$ of equal side, what is the sum of $\overrightarrow{AB}+\overrightarrow{AC}+\overrightarrow{AD}+\overrightarrow{AE}+\overrightarrow{AF}+\overrightarrow{AG}+\overrightarrow{AH}$, if, $\overrightarrow{AO}=2\hat{i}+3\hat{j}-4\hat{k}$

[25 Feb, 2021 (Shift-I)]

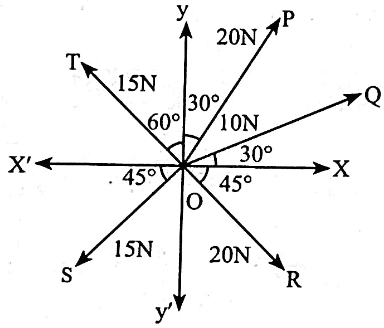


(a) $16\hat{i}+24\hat{j}-32\hat{k}$ (b) $16\hat{i}-24\hat{j}+32\hat{k}$

(c) $-16\hat{i}-24\hat{j}+32\hat{k}$ (d) $16\hat{i}+24\hat{j}-32\hat{k}$

12. The resultant of these forces $\overrightarrow{OP},\overrightarrow{OQ},\overrightarrow{OR},\overrightarrow{OS}$ and $\overrightarrow{OT}$ is approximately ...... N. **[27 Aug, 2021 (Shift-I)]**

[Take $\sqrt{3}=1.7, \sqrt{2}=1.4$. Given $\hat{i}$ and $\hat{j}$ unit vectors along x, y axis]



(a) $9.25\hat{i}+5\hat{j}$ (b) $2.5\hat{i}-14.5\hat{j}$

(c) $-1.5\hat{i}-15.5\hat{j}$ (d) $3\hat{i}+15\hat{j}$

13. **Statement-I:** If three forces $\vec{F}_1, \vec{F}_2$ and $\vec{F}_3$ are represented by three sides of a triangle and, $\vec{F}_1+\vec{F}_2=-\vec{F}_3$ then these three forces are concurrent forces and satisfy the condition for equilibrium.

Statement-II: A triangle made up of three forces $\vec{F}_1, \vec{F}_2$ and $\vec{F}_3$ as its sides taken in the same order, satisfy the condition for translatory equilibrium. In the light of the above statements, choose the most appropriate answer from the options given below:

[31 Aug, 2021 (Shift-II)]

(a) Statement-I is false but Statement-II is true

(b) Both Statement-I and Statement-II are true

(c) Statement-I is true but Statement-II is false

(d) Both Statement-I and Statement-II are false

14. Two forces having magnitude A and $\frac{A}{2}$ are perpendicular to each other. The magnitude of their resultant is **[8 April, 2023 (Shift-I)]**

(a) $\frac{\sqrt{5}A}{4}$ (b) $\frac{5A}{2}$ (c) $\frac{\sqrt{5}A^2}{2}$ (d) $\frac{\sqrt{5}A}{2}$

15. A vector in x–y plane makes an angle of 30° with y-axis. The magnitude of y-component of vector is $2\sqrt{3}$. The magnitude of x-component of the vector will be: **[15 April, 2023 (Shift-I)]**

(a) $\frac{1}{\sqrt{3}}$ (b) 6 (c) $\sqrt{3}$ (d) 2

16. A mosquito is moving with a velocity $\vec{v}=0.5t^2\hat{i}+3t\hat{j}+9\hat{k}$ m/s and accelerating in uniform conditions What will be the direction of mosquito after 2s? **[16 March, 2021 (Shift-II)]**

(a) $\tan^{-1}\left(\frac{\sqrt{85}}{6}\right)$ from y-axis (b) $\tan^{-1}\left(\frac{5}{2}\right)$ from y-axis

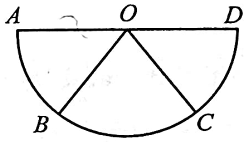
(c) $\tan^{-1}\left(\frac{2}{3}\right)$ from x-axis (d) $\tan^{-1}\left(\frac{5}{2}\right)$ from x-axis

17. **Assertion A:** If A, B, C and D are four points on a semicircular arc with centre at 'O' such that $|\overrightarrow{AB}|=|\overrightarrow{BC}|=|\overrightarrow{CD}|$, then

$\overrightarrow{AB}+\overrightarrow{AC}+\overrightarrow{AD}=4\overrightarrow{AO}+\overrightarrow{OB}+\overrightarrow{OC}$

Reason R: Polygon law of vector addition yields

$\overrightarrow{AB}+\overrightarrow{BC}+\overrightarrow{CD}=\overrightarrow{AD}=2\overrightarrow{AO}$



In the light of the above statements, choose the most appropriate answer from the options given below: **[27 July, 2021 (Shift-I)]**

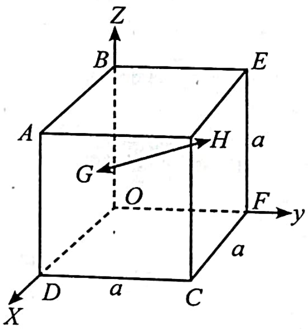
(a) **A** is not correct but **R** is correct.

(b) **A** is correct but **R** is not correct.

(c) Both **A** and **R** are correct and **R** is the correct explanation of **A**.

(d) Both **A** and **R** are correct but **R** is not the correct explanation of **A**.

18. In the cube of side 'a' shown in the figure, the vector from the central point of the face $ABOD$ to the central point of the face $BEFO$ will be **[10 Jan, 2019 (Shift-I)]**



(a) $\frac{1}{2}a(\hat{k}-\hat{i})$ (b) $\frac{1}{2}a(\hat{i}-\hat{k})$ (c) $\frac{1}{2}a(\hat{j}-\hat{i})$ (d) $\frac{1}{2}a(\hat{j}-\hat{k})$

Dot Product

19. Two particles are located at equal distance from origin. The position vectors of those are represented by $\vec{A}=2\hat{i}+3n\hat{j}+2\hat{k}$ and $\vec{B}=2\hat{i}-2\hat{j}+4p\hat{k}$, respectively. If both the vectors are at right angle to each other, the value of n^{-1} is ________.

[23 Jan, 2025 (Shift-I)]

20. If two vectors $\vec{P}=\hat{i}+2m\hat{j}+m\hat{k}$ and $\vec{Q}=4\hat{i}-2\hat{j}+m\hat{k}$ are perpendicular to each other. Then, the value of m will be:

[24 Jan, 2023 (Shift-II)]

(a) 1 (b) −1 (c) −3 (d) 2

21. Vectors $a\hat{i}+b\hat{j}+\hat{k}$ and $2\hat{i}-3\hat{j}+4\hat{k}$ are perpendicular to each other when $3a+2b=7$, the ratio of a to b is $\frac{x}{2}$. The value of x is ________.

[24 Jan, 2023 (Shift-I)]

22. Which of the following relations is true for two unit vector $\hat{A}$ and $\hat{B}$ making an angle θ to each other? **[25 June, 2022 (Shift-I)]**

(*a*) $|\hat{A}+\hat{B}|=|\hat{A}-\hat{B}|\tan\frac{\theta}{2}$ (*b*) $|\hat{A}-\hat{B}|=|\hat{A}+\hat{B}|\tan\frac{\theta}{2}$

(*c*) $|\hat{A}+\hat{B}|=|\hat{A}-\hat{B}|\cos\frac{\theta}{2}$ (*d*) $|\hat{A}-\hat{B}|=|\hat{A}+\hat{B}|\cos\frac{\theta}{2}$

23. Two vectors $\vec{A}$ and $\vec{B}$ have equal magnitudes. If magnitude of $\vec{A}+\vec{B}$ is equal to two times the magnitude of $\vec{A}-\vec{B}$, then the angle between $\vec{A}$ and $\vec{B}$ will be: **[29 June, 2022 (Shift-I)]**

(*a*) $\sin^{-1}\left(\frac{3}{5}\right)$ (*b*) $\sin^{-1}\left(\frac{1}{3}\right)$ (*c*) $\cos^{-1}\left(\frac{3}{5}\right)$ (*d*) $\cos^{-1}\left(\frac{1}{3}\right)$

24. If $\vec{A}=(2\hat{i}+3\hat{j}-\hat{k})m$ and $\vec{B}=(\hat{i}+2\hat{j}-2\hat{k})m$. The magnitude of component of vector $\vec{A}$ along vector $\vec{B}$ will be ____________ m. **[26 July, 2022 (Shift-II)]**

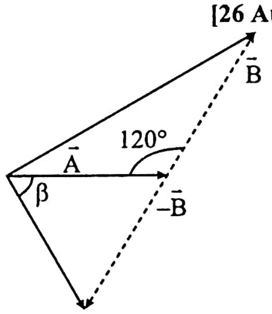
25. If the projection of $2\hat{i}+4\hat{j}-2\hat{k}$ on $\hat{i}+2\hat{j}+\alpha\hat{k}$ is zero. Then, the value of α will be _________. **[28 July, 2022 (Shift-I)]**

26. If $\vec{A}$ and $\vec{B}$ are two vectors satisfying the relation $\vec{A}.\vec{B}=|\vec{A}\times\vec{B}|$. Then the value of $|\vec{A}-\vec{B}|$ will be: **[20 July, 2021 (Shift-I)]**

(*a*) $\sqrt{A^2+B^2-\sqrt{2}AB}$ (*b*) $\sqrt{A^2+B^2}$

(*c*) $\sqrt{A^2+B^2+2AB}$ (*d*) $\sqrt{A^2+B^2+\sqrt{2}AB}$

27. The angle between vector $(\vec{A})$ and $(\vec{A}-\vec{B})$ is: **[26 Aug, 2021 (Shift-II)]**



(*a*) $\tan^{-1}\left(\frac{\sqrt{3}B}{2A-B}\right)$ (*b*) $\tan^{-1}\left(\frac{B\cos\theta}{A-B\sin\theta}\right)$

(*c*) $\tan^{-1}\left(\frac{A}{0.7\,B}\right)$ (*d*) $\tan^{-1}\left(\frac{\frac{-B}{2}}{A-B\frac{\sqrt{3}}{2}}\right)$

28. What will be the projection of vector $\vec{A}=\hat{i}+\hat{j}+\hat{k}$ on vector $\vec{B}=\hat{i}+\hat{j}$? **[22 July, 2021 (Shift-II)]**

(*a*) $2(\hat{i}+\hat{j}+\hat{k})$ (*b*) $\sqrt{2}(\hat{i}+\hat{j})$

(*c*) $(\hat{i}+\hat{j})$ (*d*) $\sqrt{2}(\hat{i}+\hat{j}+\hat{k})$

29. Two vectors $\vec{A}$ and $\vec{B}$ have equal magnitudes. The magnitude of $(\vec{A}+\vec{B})$ is '*n*' times the magnitude of $(\vec{A}-\vec{B})$. The angle between $\vec{A}$ and $\vec{B}$ is:

(*a*) $\cos^{-1}\left[\frac{n^2-1}{n^2+1}\right]$ (*b*) $\cos^{-1}\left[\frac{n-1}{n+1}\right]$

(*c*) $\sin^{-1}\left[\frac{n^2-1}{n^2+1}\right]$ (*d*) $\sin^{-1}\left[\frac{n-1}{n+1}\right]$

30. Two vectors $\vec{X}$ and $\vec{Y}$ have equal magnitude. The magnitude of $(\vec{X}-\vec{Y})$ is n times the magnitude of $(\vec{X}+\vec{Y})$. The angle between $\vec{X}$ and $\vec{Y}$ is: **[25 July, 2021 (Shift-II)]**

(*a*) $\cos^{-1}\left(\frac{n^2+1}{n^2-1}\right)$ (*b*) $\cos^{-1}\left(\frac{-n^2-1}{n^2-1}\right)$

(*c*) $\cos^{-1}\left(\frac{n^2-1}{-n^2-1}\right)$ (*d*) $\cos^{-1}\left(\frac{n^2+1}{-n^2-1}\right)$

31. Statement-I: Two forces $(\vec{P}+\vec{Q})$ and $(\vec{P}-\vec{Q})$ where $\vec{P}\perp\vec{Q}$, when act at an angle θ_1 to each other, the magnitude of their resultant is $\sqrt{3(P^2+Q^2)}$, when they act at an angle θ_2, the magnitude of their resultant becomes $\sqrt{2(P^2+Q^2)}$. This is possible only when $\theta_1<\theta_2$. **[31 Aug, 2021 (Shift-II)]**

Statement-II: In the situation given above. $\theta_1 = 60°$ and $\theta_2 = 90°$

In the light of the above statements, choose the most appropriate answer from the options given below:

(*a*) Both Statement-I and Statement-II are true
(*b*) Both Statement-I and Statement-II are false
(*c*) Statement-I is true but Statement-II is false
(*d*) Statement-I is false but Statement-II is true

32. The sum of two forces $\vec{P}$ and $\vec{Q}$ is $\vec{R}$ such that $|\vec{R}|=|\vec{P}|$. The angle θ (in degrees) that the resultant of $2\vec{P}$ and $\vec{Q}$ will make with $\vec{Q}$ is **[7 Jan, 2020 (Shift-II)]**

33. Let $|\vec{A}_1|=3, |\vec{A}_2|=5$ and $|\vec{A}_1+\vec{A}_2|=5$. The value of $(2\vec{A}_1+3\vec{A}_2).(3\vec{A}_1-2\vec{A}_2)$ is: **[8 April, 2019 (Shift-II)]**

(*a*) –112.5 (*b*) –106.5 (*c*) –118.5 (*d*) –99.5

34. Two forces P and Q, of magnitude $2F$ and $3F$, respectively, are at an angle θ with each other. If the force Q is doubled, then their resultant also gets doubled. Then, the angle θ is: **[10 Jan, 2019 (Shift-II)]**

(*a*) 120° (*b*) 60° (*c*) 90° (*d*) 30°

Cross Product

35. For three vectors $\vec{A}=(-x\hat{i}-6\hat{j}-2\hat{k})$, $\vec{B}=(-\hat{i}+4\hat{j}+3\hat{k})$ and $\vec{C}=(-8\hat{i}-\hat{j}+3\hat{k})$, if $\vec{A}\cdot(\vec{B}\times\vec{C})=0$, them value of x is _________. **[06 April, 2024 (Shift-I)]**

36. If $\vec{P}=3\hat{i}+\sqrt{3}\hat{j}+2\hat{k}$ and $\vec{Q}=4\hat{i}+\sqrt{3}\hat{j}+2.5\hat{k}$ then, the unit vector in the direction of $\vec{P}\times\vec{Q}$ is $\frac{1}{x}\left(\sqrt{3}\hat{i}+\hat{j}-2\sqrt{3}\hat{k}\right)$. The value of x is **[25 Jan, 2023 (Shift-I)]**

37. $\vec{A}$ is a vector quantity such that $|\vec{A}|$ = non-zero constant. Which of the following expression is true for $\vec{A}$? **[25 June, 2022 (Shift-I)]**

(*a*) $\vec{A}.\vec{A}=0$ (*b*) $\vec{A}\times\vec{A}<0$ (*c*) $\vec{A}\times\vec{A}=0$ (*d*) $\vec{A}\times\vec{A}>0$

38. If force $\vec{F}=3\hat{i}+4\hat{j}-2\hat{k}$ acts on a particle having position vector $2\hat{i}+\hat{j}+2\hat{k}$ then, the torque about the origin will be: **[25 June, 2022 (Shift-I)]**

(*a*) $3\hat{i}+4\hat{j}-2\hat{k}$ (*b*) $-10\hat{i}+10\hat{j}+5\hat{k}$

(*c*) $10\hat{i}+5\hat{j}-10\hat{k}$ (*b*) $10\hat{i}+\hat{j}-5\hat{k}$

39. If $\vec{P}\times\vec{Q}=\vec{Q}\times\vec{P}$ the angle between $\vec{P}$ and $\vec{Q}$ is θ ($0°<\theta<360°$). The value of 'θ' will be _______ **[25 Feb, 2021 (Shift-II)]**

Function, Differentiation as a Rate Measurement

40. A person of height 1.6 m is walking away from a lamp post of height 4m along a straight path on the flat ground. The lamp post and the person are always perpendicular to the ground. If the speed of the person is 60 cm s^{-1}, the speed of the tip of the person's shadow on the ground with respect to the person is ________ cm s^{-1}. **[JEE Adv, 2023]**

41. If $t = \sqrt{x} + 4$, then $\left(\frac{dx}{dt}\right)_{t=4}$ is: **[29 July, 2022 (Shift-I)]**

(a) 4 (b) zero (c) 8 (d) 16

42. Three particles P, Q and R are moving along the vectors $\vec{A} = \hat{i} + \hat{j}, \vec{B} = \hat{j} + \hat{k}$ and $\vec{C} = -\hat{i} + \hat{j}$ respectively. They strike on point and start to move in different directions. Now particle P is moving normal to the plane which contains vector $\vec{A}$ and $\vec{B}$. Similarly particle Q is moving normal to the plane which contains vector $\vec{A}$ and $\vec{C}$. The angle between the direction of motion of P and Q is $\cos^{-1}\left(\frac{1}{\sqrt{x}}\right)$. Then the value of x is ________. **[22 July, 2021 (Shift-II)]**

ANSWER KEY

1. (c)	2. [4]	3. [6]	4. (a)	5. [3]	6. [3]	7. [150]	8. (*)	9. (b)	10. (c)
11. (a)	12. (a)	13. (b)	14. (d)	15. (d)	16. (a)	17. (d)	18. (c)	19. [3]	20. (d)
21. [1]	22. (b)	23. (c)	24. [2]	25. [5]	26. (a)	27. (a)	28. (c)	29. (a)	30. (c)
31. (a)	32. [90°]	33. (c)	34. (a)	35. (4)	36. [4]	37. (c)	38. (b)	39. [180]	40. [40]
41. (b)	42. [3]								